Lecture 14: The Factor Zoo - Part 1

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Intro

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- Last lecture: factor models worked, but interpretation was messy
- Today: maybe empirical success is the easy part
- LNS attacks the tests
- Harvey-Liu-Zhu attack the t-stats
- Kelly-Pruitt-Su change the object

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- LNS attacks the tests
- Harvey-Liu-Zhu attack the t-stats
- Kelly-Pruitt-Su change the object
- Main theme: the data are high-dimensional, our tests are not

Flight Plan

- Part I: why Lewellen et al. (2010) changed the standards for factor tests
- Part II: how Lewellen et al. (2010) tells us to raise the bar
- Part III: how Fama and French (2015) expands the benchmark
- Part IV: why Harvey et al. (2016) turns the factor zoo into an inference problem
- Part V: how Kelly et al. (2019) moves from factors to structure

Part I: We Should Have Higher Standards...

The Factor-Zoo Explosion

During the 2000s and early 2010s:

- Many strategies with large expected returns and no relation to betas appeared
- Examples: accruals, profitability, investment, asset growth, ...
- See Fama and French (2008) for an extensive analysis

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- Also: several papers proposing theories + factors that price the 25 portfolios
- Example: Lettau and Ludvigson (2001)

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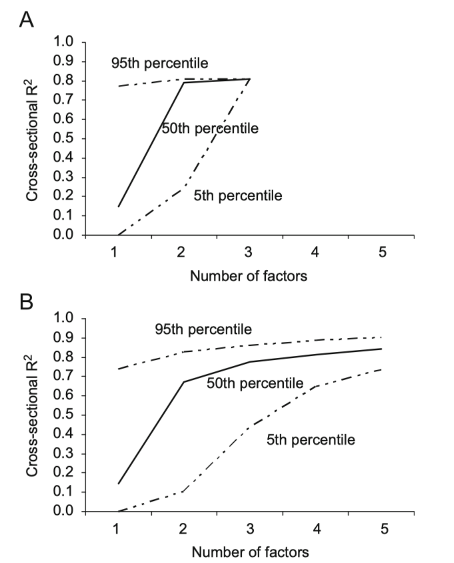
- Examples: accruals, profitability, investment, asset growth, ...
- See Fama and French (2008) for an extensive analysis
- Also: several papers proposing theories + factors that price the 25 portfolios
- Example: Lettau and Ludvigson (2001)
- Lewellen et al. (2010): pricing the 25 portfolios is too easy!

Lewellen-Nagel-Shanken: The Deeper Problem

- The 25 portfolios have a very strong factor structure
- We already know that MKT, SMB, and HML price them well
- If you give me a factor that is correlated with them… it will price the 25 portfolios too!
- We're judging models based on just a few assets!

A Killer Simulation



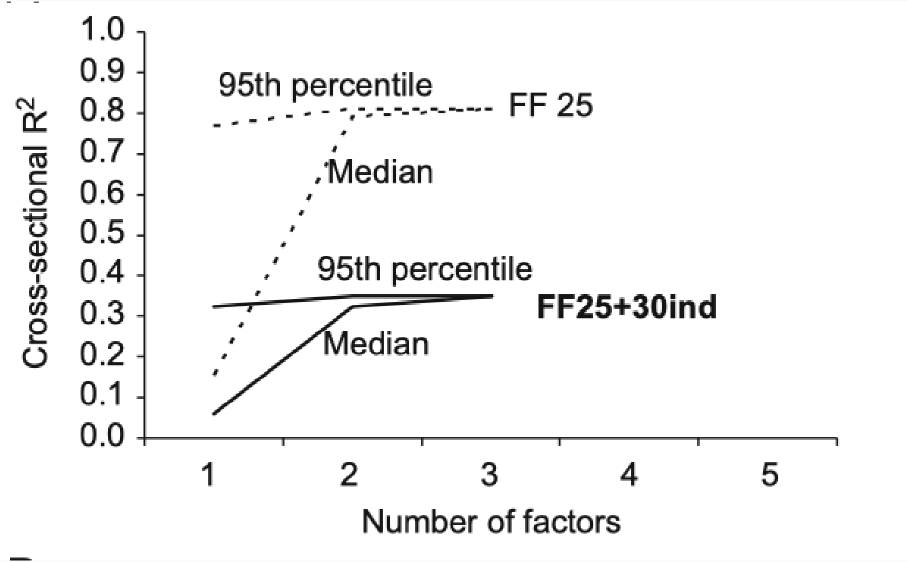
- Panel A: simulate random combinations of MKT, SMB, and HML
- Estimate the time-series betas and then run the cross-sectional pass
- Repeat many times and keep track of R^2
- Panel B: do the same, but with combinations of the 25 portfolios
- If you're not scared, look again

How To Improve Tests

They provide a few ways to improve the tests:

1. Expand the set of test portfolios beyond size-B/M portfolios
2. Take the magnitude of the cross-sectional slopes seriously
3. Report the GLS R^2 instead of OLS R^2
4. If a factor is traded, it should price itself (i.e., have a significant risk premium)
5. Report confidence intervals for the cross-sectional R^2

One Example of Improvement



Questions?

Part III: Fama-French Reloaded

The Profitability and Investment Anomalies

- Titman et al. (2004): firms that invest more heavily have **lower** subsequent returns
- This is true controlling for size, B/M, and momentum
- Novy-Marx (2013): firms with higher profitability have **higher** subsequent returns
- The 3-factor model cannot explain these anomalies
- Fama and French (2006) had already noticed these two anomalies, with different implementations

This Actually Has Some Interpretation

Recall the basic pricing formula:

$$P_t = \sum_{i=1}^{\infty} \mathbb{E}_t \left[\frac{D_{t+i}}{(1+r)^{t+i}} \right]$$

- A firm with earnings Y_{t+j} that pays dividends D_{t+j} has book equity $B_{t+j} = B_{t+j-1} + Y_{t+j} - D_{t+j}$
- The valuation formula becomes:

$$P_t = \sum_{i=1}^{\infty} \mathbb{E}_t \left[\frac{Y_{t+i} - (B_{t+i} - B_{t+i-1})}{(1+r)^{t+i}} \right]$$

- Retained earnings (changes in book equity) are typically used for investment
- Equipment, R&D, marketing, etc...

FF (2015): Extending the Benchmark

- These two anomalies were well-known and hard to explain with the 3-factor model
- Fama and French (2015): add two more factors to capture profitability and investment
- This is a direct extension of Fama and French (1993)... and a certain "concession"

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- These two anomalies were well-known and hard to explain with the 3-factor model
- Fama and French (2015): add two more factors to capture profitability and investment
- This is a direct extension of Fama and French (1993)... and a certain “concession”
- Two new factors created similarly to SMB and HML, but sorting on profitability and investment
- Profitability: *robust minus weak* (RMW)
- Investment: *conservative minus aggressive* (CMA)

The Five Factors

- MKT, SMB, and HML are measured as in Fama and French (1993)
- Operating profitability: think about EBITDA
- Investment: asset growth

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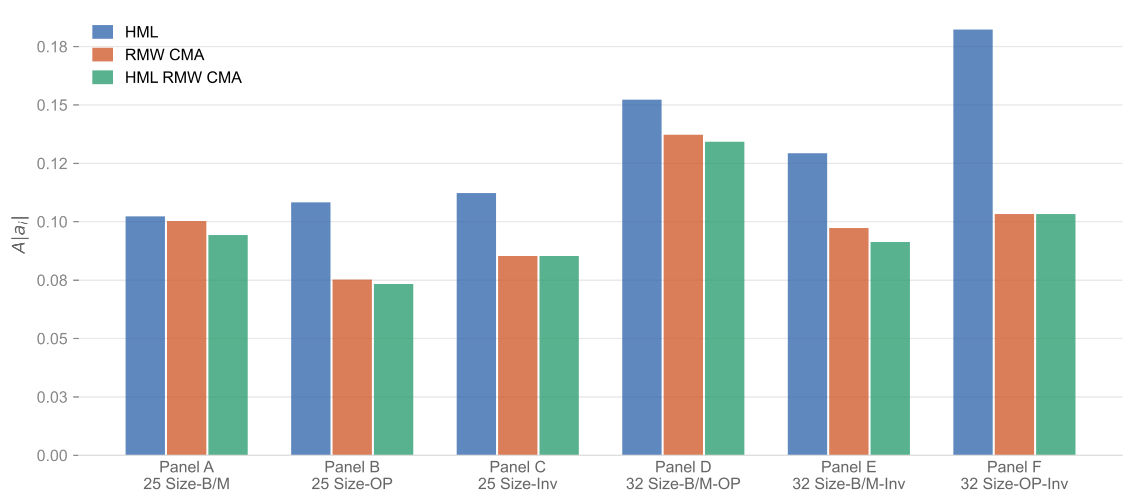
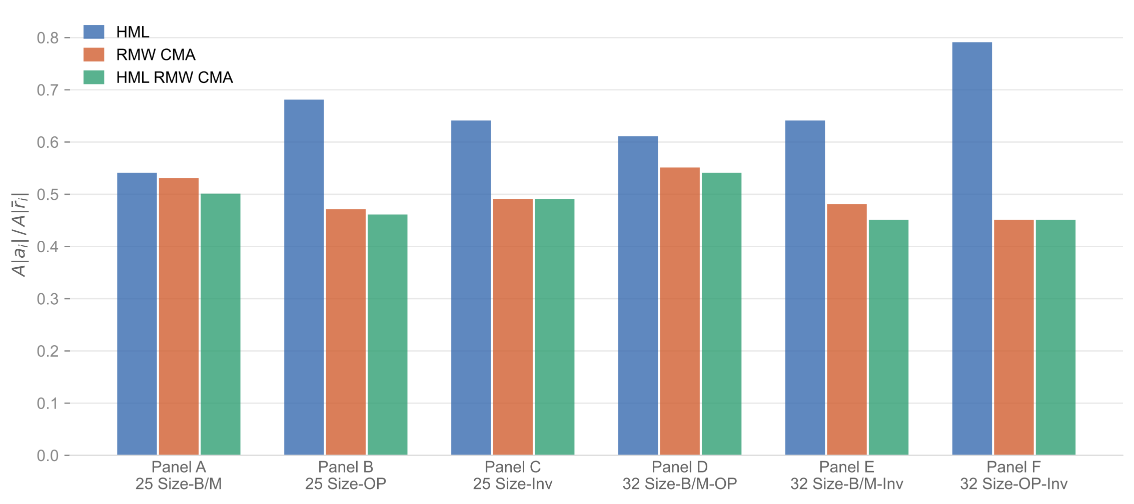
- MKT, SMB, and HML are measured as in Fama and French (1993)
- Operating profitability: think about EBITDA
- Investment: asset growth
- Create 2x2 sorts of (size, other variable), where other variable = B/M, profitability, or investment
- They study many ways to construct RMW and CMA and results won't change
- RMW and CMA are just zero-cost long-short portfolios, like SMB and HML

FF (2015): Mean Absolute Intercepts



FF (2015): Mean Absolute Intercepts / Mean Absolute Returns



Fama-French (2015): Two Main Empirical Lessons

- Adding these factors did decrease the average α
- HML becomes somewhat redundant
- Vast majority of cases: the model fails the GRS test from Gibbons et al. (1989)
- What does that mean?

Fama-French (2015): Two Main Empirical Lessons

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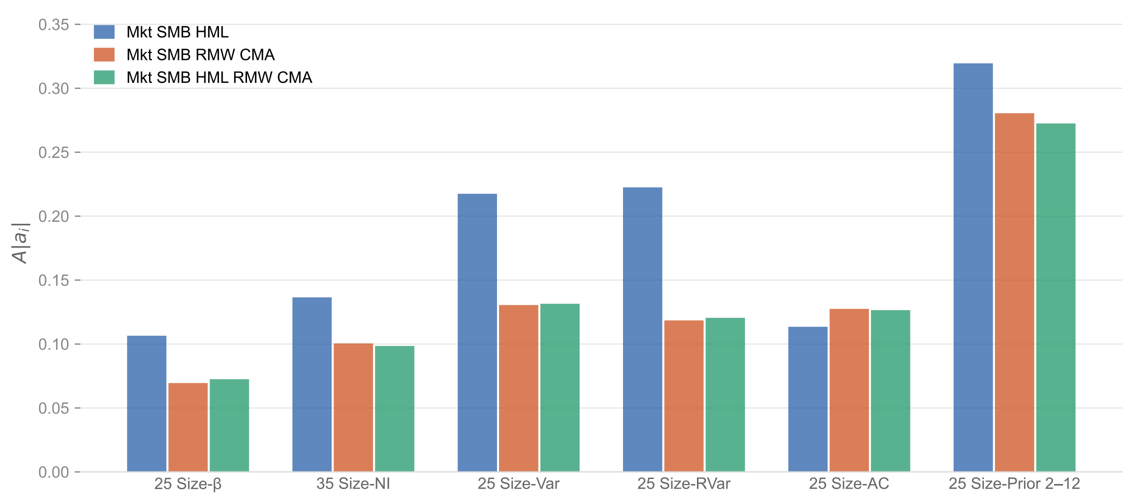
> Asset pricing models are simplified propositions about expected returns that are rejected in tests with power. We are less interested in whether competing models are rejected than in their relative performance, which we judge using *GRS* and other statistics. We want to identify the model that is the best (but imperfect) story for average returns on portfolios formed in different ways.

Questions?

Fama-French (2016): Dissecting Anomalies

- Fama and French (2016) took Lewellen et al. (2010) more seriously
- Unlike Fama and French (2015), they try to fit returns of portfolios sorted on different characteristics
- Examples: size-beta, size-net share issues, size-volatility, size-accruals, size-momentum
- Same vibe: create several sorted portfolios, run regressions, check intercepts and fit

FF (2016): Mean Absolute Intercepts



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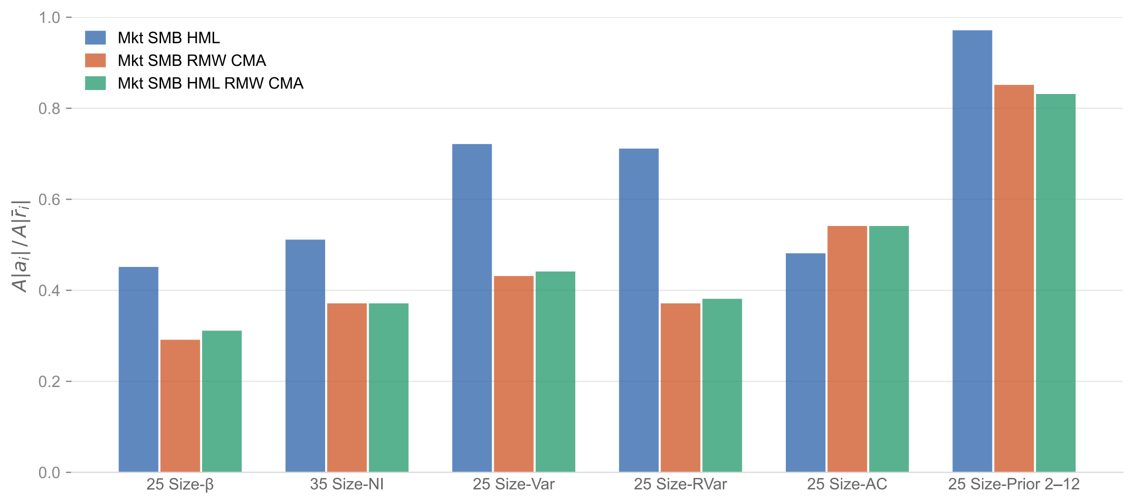


Empirical Lessons

- The 5-factor model is useful in the sense it reduces intercepts
- It essentially never passes the GRS test, though
- Two main enemies: momentum and, to some extent, accruals
- Another cool thing: they explain why the security market liner appeared flatter
- Reason: stocks with low CAPM betas load heavily on HML, RMW, and CMA
- Stocks with high CAPM betas also load heavily on factors, with a *negative* coefficient

Questions?

Part IV: The Factor Zoo

Harvey, Liu, and Zhu (2016)

- Harvey et al. (2016) collect 316 factors, from 313 papers published since the 60s in top journals
- The first paper I know of that tried to "catalog" the factor zoo
- Multiple testing is the core problem
- Even if you generate random factors, some of them will the price the cross-section by chance
- If you keep doing that forever, you'll find many "significant" factors
- Their prescription: increase the hurdle for statistical significance
- The "$t > 1.96$" rule-of-thumb is not enough
- By the way: t here might mean the t-stat of either the CS or the TS regression

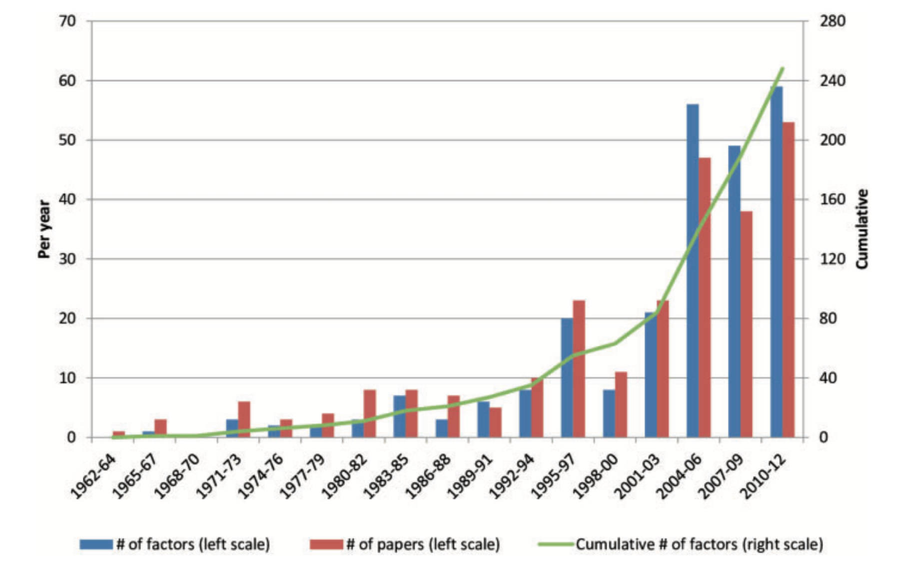
Taxonomy of Factors

Table 1
Factor classification

Risk type		Description	Examples
Common (113)	**Financial** (46)	Proxy for aggregate financial market movement, including market portfolio returns, volatility, squared market returns, among others	Sharpe (1964): market returns; Kraus and Litzenberger (1976): squared market returns
	Macro (40)	Proxy for movement in macroeconomic fundamentals, including consumption, investment, inflation, among others	Breeden (1979): consumption growth; Cochrane (1991): investment returns
	Microstructure (11)	Proxy for aggregate movements in market microstructure or financial market frictions, including liquidity, transaction costs, among others	Pastor and Stambaugh (2003): market liquidity; Lo and Wang (2006): market trading volume
	Behavioral (3)	Proxy for aggregate movements in investor behavior, sentiment or behavior-driven systematic mispricing	Baker and Wurgler (2006): investor sentiment; Hirshleifer and Jiang (2010): market mispricing
	Accounting (8)	Proxy for aggregate movement in firm-level accounting variables, including payout yield, cash flow, among others	Fama and French (1992): size and book-to-market; Da and Warachka (2009): cash flow
	Other (5)	Proxy for aggregate movements that do not fall into the above categories, including momentum, investors' beliefs, among others	Carhart (1997): return momentum; Ozoguz (2009): investors' beliefs
Characteristics (202)	**Financial** (61)	Proxy for firm-level idiosyncratic financial risks, including volatility, extreme returns, among others	Ang et al. (2006): idiosyncratic volatility; Bali, Cakici, and Whitelaw (2011): extreme stock returns
	Microstructure (28)	Proxy for firm-level financial market frictions, including short sale restrictions, transaction costs, among others	Jarrow (1980): short sale restrictions; Mayshar (1981): transaction costs
	Behavioral (3)	Proxy for firm-level behavioral biases, including analyst dispersion, media coverage, among others	Diether, Malloy, and Scherbina (2002): analyst dispersion; Fang and Peress (2009): media coverage
	Accounting (87)	Proxy for firm-level accounting variables, including PE ratio, debt-to-equity ratio, among others	Basu (1977): PE ratio; Bhandari (1988): debt-to-equity ratio
	Other (24)	Proxy for firm-level variables that do not fall into the above categories, including political campaign contributions, ranking-related firm intangibles, among others	Cooper, Gulen, and Ovtchinnikov (2010): political campaign contributions; Edmans (2011): intangibles

The numbers in parentheses represent the number of factors identified. See Table 6 and http://faculty.fuqua.duke.edu/~charvey/Factor-List.xlsx.

Factors Over Time



The Intuition From Bonferroni

- If you test 20 factors, the probability of at least one false positive is $1 - (1 - 0.05)^{20} \approx 0.64$
- With 100 factors, the probability of at least one false positive is $1 - (1 - 0.05)^{100} \approx 0.994$

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- With 100 factors, the probability of at least one false positive is $1-(1-0.05)^{100} \approx 0.994$
- Suppose you test M hypotheses with tests of size α
- Let A_i be the event that the i-th test is a rejection when the null is true

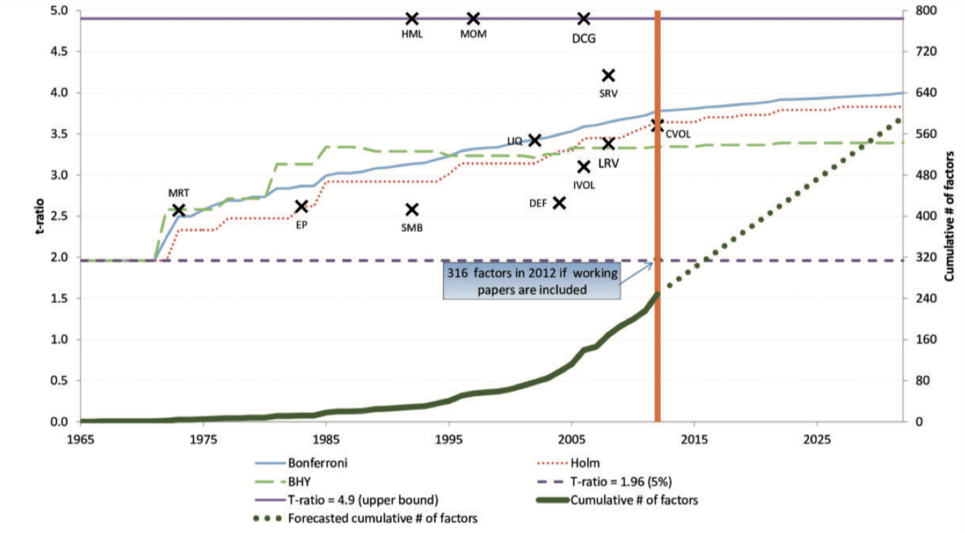
$$\mathbb{P}\left(\cup_{i=1}^{M} A_i\right) \leq \sum_{i=1}^{M} \mathbb{P}(A_i) = M\alpha$$

- The "Bonferroni correction" says: reject if the p-value is less than α/M
- Lower p-values imply higher thresholds for t-statistics

What They Do

- At each date, count the cumulative number of published factors, M_t
- Take the available factor t-statistics and convert them into p-values
- Bonferroni: use the cutoff $p_t^\star = \alpha / M_t$
- Convert $p_t^\star$ back into an implied threshold $t_t^\star$
- Trace this hurdle through time as the zoo grows
- Same exercise for Holm and BHY, which are other corrections

Empirical Result



Why Should It Increase Over Time?

- Their logic is about the search process, not the null
- Low-hanging fruit gets picked first
- The data set is still basically finite
- Computing and data got much cheaper
- For future dates, they linearly extrapolate factor production using 2003–2012
- So a new factor today should clear a higher hurdle

Main Empirical Lessons

- The usual $t > 1.96$ standard is too low in the factor zoo
- Once multiple testing is accounted for, the cutoff is closer to $t \approx 3$
- Many published factors no longer clear Bonferroni, Holm, or BHY
- In their final count, about 27%–53% of published significant factors are likely false discoveries
- If we also account for unpublished searches, the hurdle is even higher
- Limitation: the true number of attempted factors is unobserved
- Limitation: factor tests are correlated, so standard corrections may be too harsh
- Super influential paper: many factors are just a spark of luck!

Questions?

Part V: Characteristics Are Covariances

Motivation for Kelly, Pruitt, and Su (2019)

- The true challenge in asset pricing: loadings and factors are latent
- Potentially time-varying too!

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Motivation for Kelly, Pruitt, and Su (2019)

- The true challenge in asset pricing: loadings and factors are latent
- Potentially time-varying too!
- PCA-based techniques (Chamberlain and Rothschild 1983) help, but loadings are constant
- Factor zoo: we probably have redundant information...
- Different combinations two factors should not be new factor
- The linear span of factors is what really matters

What They Do

- Provide a flexible asset pricing model with many desirable features
- Factors are latent, so the researcher does not specify them ex ante
- Loadings are time-varying and are *functions* of characteristics (Ferson and Harvey 1999)
- Covariances (betas) *are* characteristics (Daniel and Titman 1997)
- They can easily apply the model to stocks directly, not just portfolios (Lewellen et al. 2010)
- The model is parsimonious: fewer parameters than, for example, FF5

Are Characteristics Generating Covariances? Or Just Mispricing?

- In a rational asset-pricing model, characteristics should not earn returns "by themselves"
- They should matter because they forecast conditional exposures to common risks

$$\mathbb{E}_t[r_{i,t+1}] = \alpha_{i,t} + \beta_{i,t}^\top \lambda_t$$

- The question is where characteristics belong: in $\beta_{i,t}$ or in $\alpha_{i,t}$
- Their answer: use characteristics to instrument conditional betas on latent factors
- Assumption/limitation: λ_t is constant (and estimated)

(Instrumented)PCA

$$r_{i,t+1} = \alpha_{i,t} + \beta_{i,t}^\top f_{t+1} + \varepsilon_{i,t+1}, \qquad \alpha_{i,t} = z_{i,t}^\top \Gamma_\alpha + \nu_{i,\alpha,t}, \qquad \beta_{i,t} = z_{i,t}^\top \Gamma_\beta + \nu_{i,\beta,t}$$

- $z_{i,t}$: lagged firm characteristics, including a constant
- f_{t+1}: latent factors, to be estimated
- $\Gamma_\alpha, \Gamma_\beta$: mapping from characteristics to expected returns and factor loadings

(Instrumented)PCA

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- $z_{i,t}$: lagged firm characteristics, including a constant
- f_{t+1}: latent factors, to be estimated
- $\Gamma_{\alpha}, \Gamma_{\beta}$: mapping from characteristics to expected returns and factor loadings
- PCA uses only returns
- Fama-French fixes factors ex ante
- IPCA estimates latent factors, but disciplines their loadings with characteristics

Estimating Parameters

Substitute the equation for $\beta_{i,t}$ and α_{it} and write in matrix form:

$$r_{t+1} = Z_t\Gamma_\alpha + Z_t\Gamma_\beta f_{t+1} + \varepsilon^*_{t+1}$$

$$(\hat{\Gamma}_\alpha, \hat{\Gamma}_\beta, \hat{F}) = \arg\min_{\Gamma_\alpha,\Gamma_\beta,F} \sum_{t=1}^{T-1} \left(r_{t+1} - Z_t\Gamma_\alpha - Z_t\Gamma_\beta f_{t+1}\right)^\top \left(r_{t+1} - Z_t\Gamma_\alpha - Z_t\Gamma_\beta f_{t+1}\right)$$

- Γ_α collects characteristic-linked expected returns not absorbed by common factors
- Γ_β maps characteristics to factor loadings
- So IPCA nests the risk story and the anomaly story in one system

Restricted Estimator

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$$\left(\hat{\Gamma}_\beta, f_1, f_2, \dots, f_T\right) = \arg \min_{\Gamma_\beta, \{f_{t+1}\}_{t=0}^{T-1}} \sum_{t=1}^{T-1} \left(r_{t+1} - Z_t \Gamma_\beta f_{t+1}\right)^\top \left(r_{t+1} - Z_t \Gamma_\beta f_{t+1}\right)$$

The FOC for the Restricted Problem

$$\hat{f}_{t+1} = \left(\hat{\Gamma}'_\beta Z'_t Z_t \hat{\Gamma}_\beta\right)^{-1} \hat{\Gamma}'_\beta Z'_t r_{t+1}, \qquad \forall t$$

and

$$\text{vec}(\hat{\Gamma}_\beta) = \left(\sum_{t=1}^{T-1} Z'_t Z_t \otimes \hat{f}_{t+1}\hat{f}'_{t+1}\right)^{-1} \left(\sum_{t=1}^{T-1} \left[Z_t \otimes \hat{f}_{t+1}\right]' r_{t+1}\right)$$

- Factors resemble Fama and MacBeth (1973)!
- (The vec of) Γ_β is a the slope coefficient of a regression of returns on interactions
- Key contribution of their companion paper: how to estimate this!
- Similar math when Γ_α is included: see the paper!
- Python implementation: (Click here)

Wald-Type Test for Alpha

$$H_0 : \Gamma_\alpha = 0 \qquad \text{vs.} \qquad H_1 : \Gamma_\alpha \neq 0$$

$$W_\alpha = \hat{\Gamma}_\alpha^\top \hat{\Gamma}_\alpha$$

- (Wild parametric) Bootstrap under the null to get the p-value
- Fail to reject: characteristics explain returns through common exposures
- Reject: some systematic characteristic-linked alpha remains
- Similar tests for Γ_β. Why is testing Γ_β interesting?

Implementation Details

- They need to choose the number K of factors: they try $K = 1, 2, 3, 4, 5, 6$
- They have 36 characteristics: size, assets, profitability, momentum, ...
- Total of 12,813 firms, across 1964-2014
- 1,403,544 stock-month observations
- This is much higher scale than anything before them
- Instead of asking "can a certain model price this anomaly?", they ask: "can any common factor structure price all of these anomalies?"

Questions?

Performance Metrics

$$\text{Total } R^2 = 1 - \frac{\sum_{i,t}\left(r_{i,t+1} - z_{i,t}^\top\left(\hat{\Gamma}_\alpha + \hat{\Gamma}_\beta \hat{f}_{t+1}\right)\right)^2}{\sum_{i,t} r_{i,t+1}^2}$$

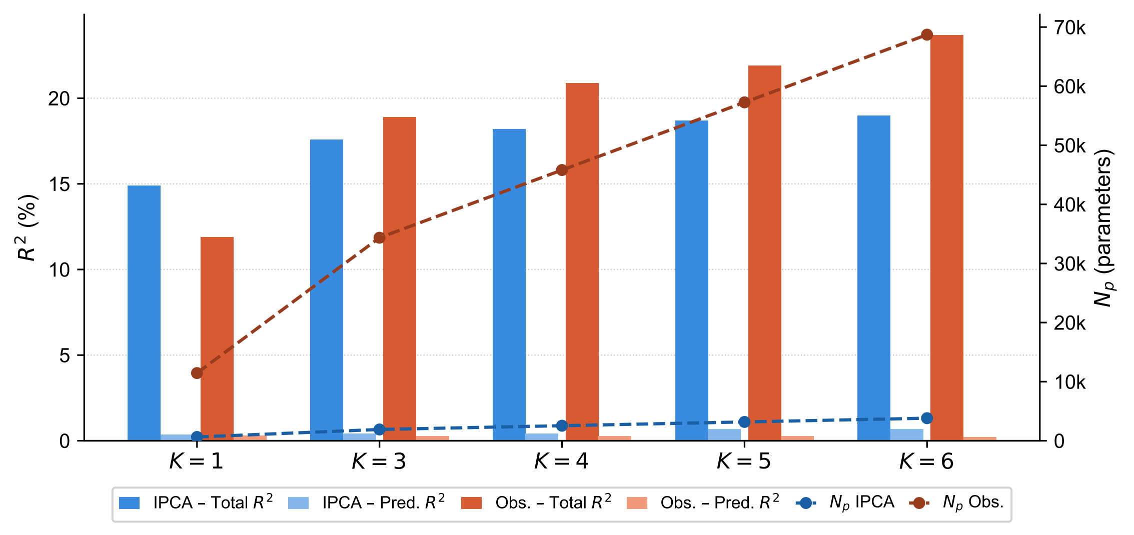
$$\text{Predictive } R^2 = 1 - \frac{\sum_{i,t}\left(r_{i,t+1} - z_{i,t}^\top\left(\hat{\Gamma}_\alpha + \hat{\Gamma}_\beta \hat{\lambda}\right)\right)^2}{\sum_{i,t} r_{i,t+1}^2}$$

- Total R^2: how much realized return variation the model explains
- This is the paper's measure of how well the model describes systematic risk in the panel
- Predictive R^2: how much realized return variation is explained by model-implied expected returns
- They hold risk prices fixed at $\hat{\lambda}$: predictive content comes from the instrumented loadings

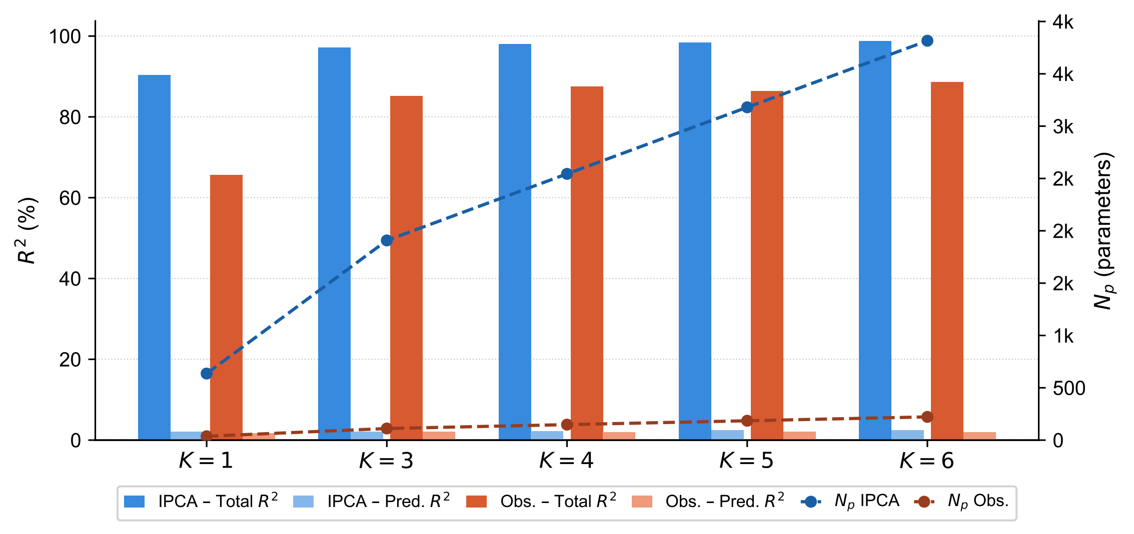
Model Performance Across K

		K					
		1	2	3	4	5	6
Panel A: Individual stocks (r_t)							
Total R^2	$\Gamma_\alpha = 0$	14.8	16.4	17.4	18.0	18.6	18.9
	$\Gamma_\alpha \neq 0$	15.2	16.8	17.7	18.4	18.7	19.0
Pred. R^2	$\Gamma_\alpha = 0$	0.35	0.34	0.41	0.42	0.69	0.68
	$\Gamma_\alpha \neq 0$	0.76	0.75	0.75	0.74	0.74	0.72
Panel B: Managed portfolios (x_t)							
Total R^2	$\Gamma_\alpha = 0$	90.3	95.3	97.1	98.0	98.4	98.8
	$\Gamma_\alpha \neq 0$	90.8	95.7	97.3	98.2	98.6	98.9
Pred. R^2	$\Gamma_\alpha = 0$	2.01	2.00	2.10	2.13	2.41	2.39
	$\Gamma_\alpha \neq 0$	2.61	2.56	2.54	2.51	2.50	2.46
Panel C: Asset pricing test							
W_α *p-value*		0.00	0.00	0.00	0.00	2.06	52.1

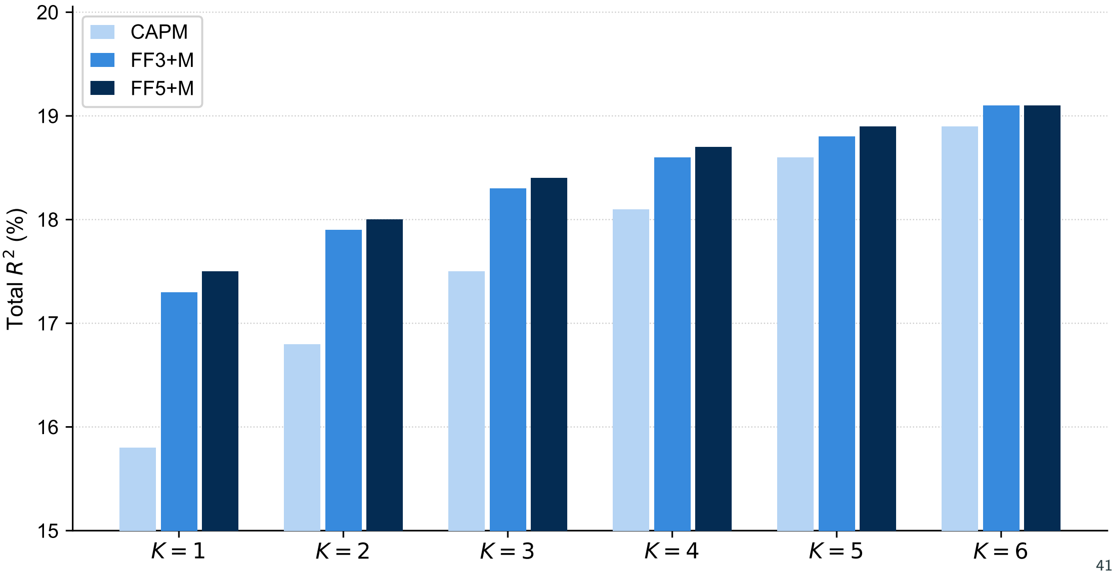
Comparison With Observable-Factor Models (Individual Stocks)



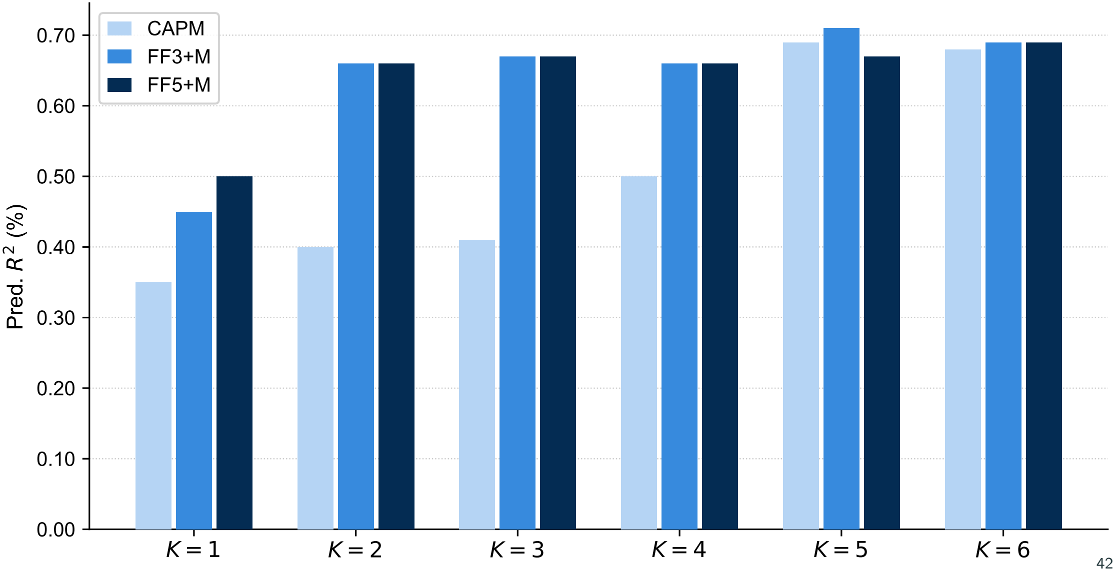
Comparison With Observable-Factor Models (Managed Portfolios)



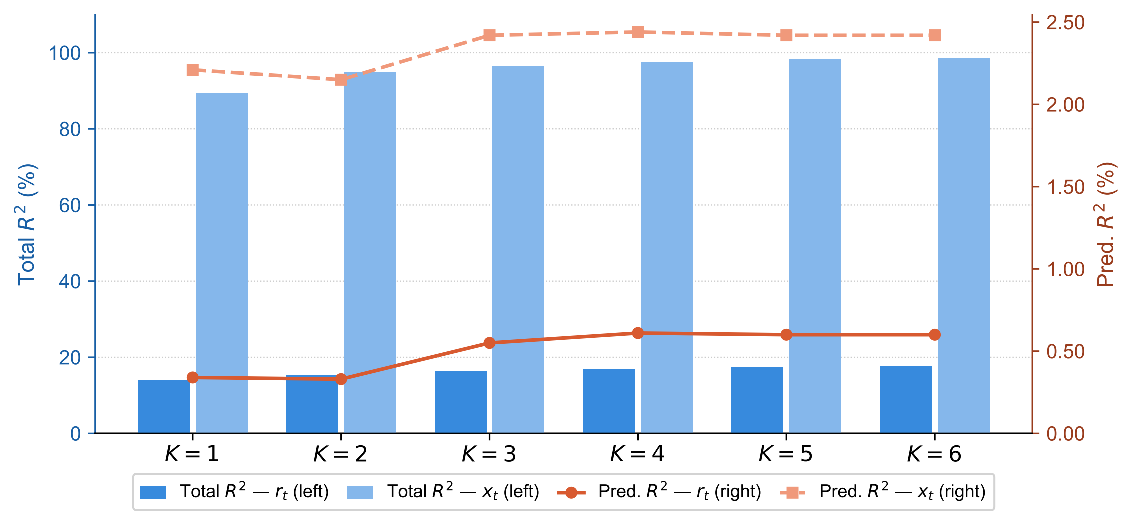
Adding Standard Factors - Total R^2



Adding Standard Factors - Predictive R^2



Out of Sample

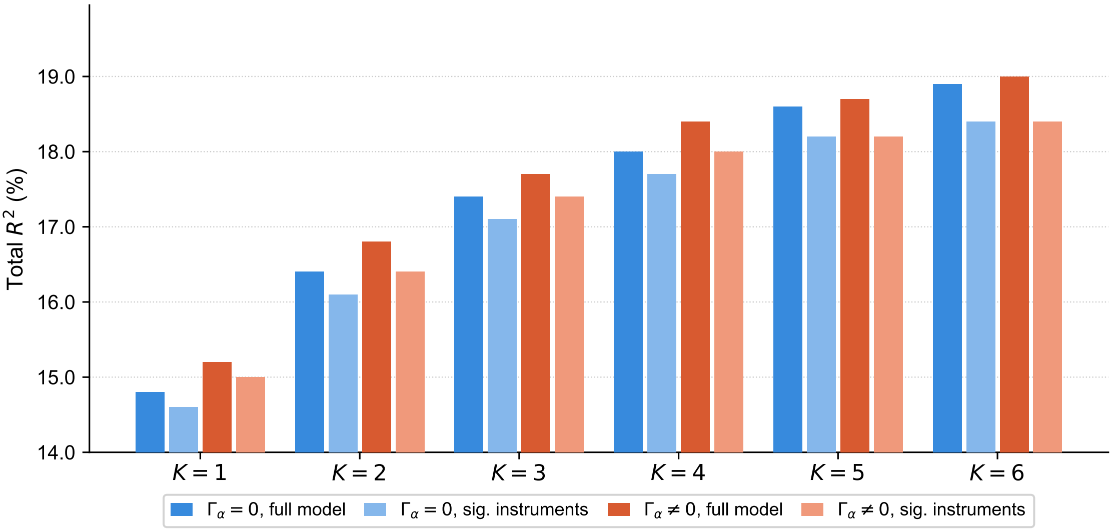


Contribution of Total R^2 For Each Variable (with $K = 5$)

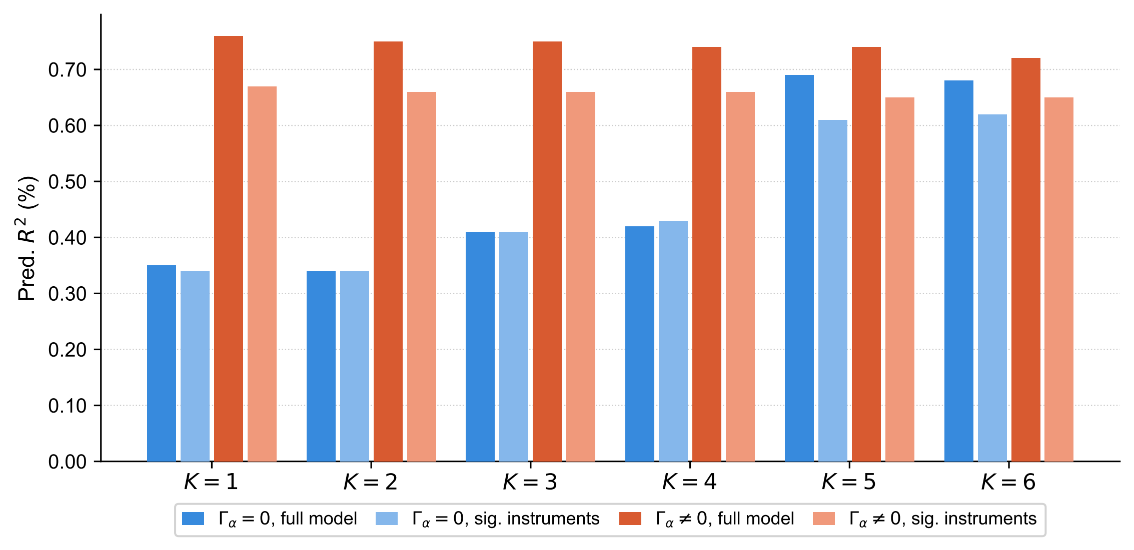
mktcap	2.84**	roa	0.07	c	0.03
assets	1.64**	suv	0.07**	noa	0.03
beta	0.56**	pcm	0.06*	rna	0.02
strev	0.47**	idiovol	0.05**	invest	0.02
mom	0.33**	s2p	0.05	prof	0.02
turn	0.31**	bm	0.04	pm	0.02
w52h	0.14**	bidask	0.04	d2a	0.01
a2me	0.14	intmom	0.03*	dpi2a	0.01
cto	0.13	roe	0.03	q	0.01
ol	0.11	sga2m	0.03	freecf	0.01
ltrev	0.10**	ato	0.03*	lev	0.01
fc2y	0.08	e2p	0.03	oa	0.00

- Only 10/36 characteristics have significant Γ_β coefficients at 5%

What Happens If You Take Out Insignificant Characteristics? - Total R^2



What Happens If You Take Out Insignificant Characteristics? – Predictive R^2



Limitations and Impact

- The factors are statistical objects; economic interpretation is still incomplete
- Results depend on the chosen characteristic set and on the linear mapping $\beta_{i,t} = \Gamma_\beta^\top z_{i,t}$
- Impact: it bridges Daniel-Titman, conditional beta models, and the modern anomaly literature
- It changed the question from "which observable factor kills alpha?" to "does any common risk structure kill alpha?"

Questions?

Readings and References i

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